Common ions and formulae of ionic compounds

Symbols of common ions

You are not expected to know all the names and symbols of common ions, but you should be able to work out the formulae of ionic compounds. The names and symbols of some ions are shown below.

Positive ions (cations)		Negative ions (anions)	
Name	Symbol	Name	Symbol
hydrogen	H^+	chloride	Cl^-
sodium	Na^+	bromide	Br^-
silver	Ag^+	fluoride	F^-
potassium	K^+	iodide	I^-
lithium	Li^+	hydrogencarbonate	HCO_3^-
ammonium	NH_4^+	hydroxide	OH^-
barium	Ba^{2+}	nitrate	NO_3^-
calcium	Ca^{2+}	oxide	O^{2-}
copper(II)	Cu^{2+}	sulphide	S^{2-}
magnesium	Mg^{2+}	sulphate	SO_4^{2-}
zinc	Zn^{2+}	carbonate	CO_3^{2-}
lead	Pb^{2+}		
iron(II)	Fe^{2+}		
iron(III)	Fe^{3+}		
aluminium	Al^{3+}		

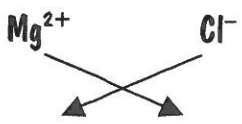
Formulae of ionic compounds

Ionic compounds contain positive and negative ions. The number of positive charges must equal the number of negative charges so that the compound has no charge overall. When the positive ion has the same number of charges as the negative ion, it is easy to work out the formula of the compound formed. Sodium chloride contains sodium ions, Na^+, and chloride ions, Cl^-. As both ions have single charges, the formula is simply written as NaCl, i.e. the positive ion followed by the negative ion with no charges written. Similarly, ammonium chloride is NH_4Cl; magnesium oxide is MgO, and so on. The fun starts when the number of charges is different, as in magnesium chloride. The "cross-over" method may help:

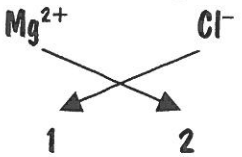
Step 1 Write the ions side by side:

Mg^{2+} Cl^-

Step 2 Draw arrows that cross each other:



Step 3 Write the **charges** at the arrow ends:



Step 4 Write the formula as follows:

a) write the positive ion without its charge:

Mg

b) write the number as a subscript unless it is 1:

Mg

c) write the negative ion without its charge:

$MgCl$

d) write the number as a subscript unless it is 1:

$MgCl_2$ … all done!

Watch out for **compound ions**, e.g. ammonium, hydrogencarbonate, hydroxide, nitrate, sulphate and carbonate. If you need more than one of them to balance the charges, put brackets around their symbol at step (a) or (c). For example, sodium hydroxide is NaOH, but magnesium hydroxide is $Mg(OH)_2$; copper(II) sulphate is $CuSO_4$, but ammonium sulphate is $(NH_4)_2SO_4$.

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NAMING AND WRITING FORMULAS FOR IONIC COMPOUNDS

EXERCISE 1

For each of the following ionic compounds, **identify the cation and the anion** (by symbol and by name of the compound.

FORMULA	CATION	ANION	NAME
KCl	K^+ Potassium	Cl^- Chloride	Potassium chloride
NH_4OH			
$ZnCl_2$			
$Al(OH)_3$			
$CuSO_4$			

EXERCISE 2

Assume that A^+, B^{2+}, C^{3+}, DE^{2+} and FG_4^+ are five cations and V^-, W^{2-}, X^{3-} and YZ^- are four anions. Write the formulas of the compounds that would result, if each of these ions were paired with one another. Complete the table below.

	V^-	W^{2-}	X^{3-}	YZ^-
A^+				
B^{2+}				
C^{3+}				
DE^{2+}				
FG_4^+				

EXERCISE 3

Write the ionic formulas for the following compounds in the spaces provided.

Potassium bromide _____

Sodium hydroxide _____

Calcium chloride _____

Calcium hydroxide _____

Potassium nitrate _____

Calcium nitrate _____

Chemical equations

Write balanced symbol equations for the following chemical reactions.
Remember to balance the equations, and to include the state symbols.

1. barium chloride + sodium sulphate → barium sulphate + sodium chloride
2. calcium + oxygen → calcium oxide
3. calcium carbonate + sulphuric acid → calcium sulphate + carbon dioxide + water
   H_2SO_4 (aq)
4. carbon + oxygen → carbon dioxide
5. copper + oxygen → copper(II) oxide
6. copper(II) carbonate + hydrochloric acid → copper(II) chloride + carbon dioxide + water
   HCl (aq)
7. copper(II) oxide + sulphuric acid → copper(II) sulphate + water
8. hydrogen + chlorine → hydrogen chloride
9. hydrogen + lead(II) oxide → lead + steam
10. hydrogen + oxygen → steam
11. iron + chlorine → iron(III) chloride
12. iron(III) chloride + ammonium hydroxide → iron(III) hydroxide + ammonium chloride
13. magnesium + steam → hydrogen + magnesium oxide
14. magnesium + sulphuric acid → hydrogen + magnesium sulphate
15. sodium + oxygen → sodium oxide
16. sodium hydroxide + copper(II) sulphate → sodium sulphate + copper(II) hydroxide
17. sodium hydroxide + phosphoric acid → sodium phosphate + water
    H_3PO_4 (aq)
18. sodium hydroxide + sulphuric acid → sodium sulphate + water
19. zinc + copper(II) sulphate → copper + zinc sulphate
20. zinc + hydrochloric acid → zinc chloride + hydrogen

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Chemical Equations *Answers*

1. $BaCl_2$ (aq) + Na_2SO_4 (aq) → $BaSO_4$ (s) + 2NaCl (aq)
2. 2Ca (s) + O_2 (g) → 2CaO (s)
3. $CaCO_3$ (s) + H_2SO_4 (aq) → $CaSO_4$ (s) + CO_2 (g) + H_2O (l)
4. C (s) + O_2 (g) → CO_2 (g)
5. 2Cu + O_2 (g) → 2CuO (s)
6. $CuCO_3$ (s) + 2HCl (aq) → $CuCl_2$ (aq) + CO_2 (g) + H_2O (l)
7. CuO (s) + H_2SO_4 (aq) → $CuSO_4$ (aq) + H_2O (l)
8. H_2 (g) + Cl_2 (g) → 2HCl (g)
9. H_2 (g) + PbO (s) → Pb (s) + H_2O (g)
10. 2H_2 (g) + O_2 (g) → 2H_2O (g)
11. 2Fe (s) +3Cl_2 (g) → 2$FeCl_3$ (g)
12. $FeCl_3$ (aq) + 3NH_4OH (aq) → $Fe(OH)_3$ (s) + 3NH_4Cl (aq)
13. Mg (s) + H_2O (g) → H_2 (g) + MgO (s)
14. Mg (s) + H_2SO_4 (aq) → H_2 (g) + $MgSO_4$ (s)
15. 4Na (s) + O_2 (g) → 2Na_2O (s)
16. 2NaOH (aq) + $CuSO_4$ (aq) → Na_2SO_4 (aq) + $Cu(OH)_2$ (aq)
17. 3NaOH (aq) + H_3PO_4 (aq) → Na_3PO_4 (aq) + 3H_2O (l)
18. 2NaOH (aq) + H_2SO_4 (aq) → Na_2SO_4 (aq) + 2H_2O (l)
19. Zn (s) + $CuSO_4$ (aq) → Cu (s) + $ZnSO_4$ (aq)
20. Zn (s) + 2HCl (aq) → $ZnCl_2$ (aq) + H_2 (g)

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Name: _____

Period: _____

Naming and Covalent Compounds

Naming

	Type: Ionic; Covalent; Polyatomic.	Compound Name
1. $MgCl_2$	*Ionic*	*Magnesium chloride*
2. PF_3		
3. CaO		
4. $K_3(PO_4)_2$		
5. $MgCl_2$		
6. CO		
7. S_2O_4		
8. $Mg(CrO_4)$		
9. NaF		
10. H_2O		
11. CO_2		
12. OBr_2		

Use only for Polyatomic Compounds

Polyatomic Ions		
Oxidation #	**Name**	**Formula**
1+	ammonium	NH_4^{+}
1-	acetate	$C_2H_3O_2^{1-}$
2-	carbonate	CO_3^{2-}
2-	chromate	CrO_4^{2-}
1-	hydrogen carbonate	HCO_3^{1-}
1+	hydronium	H_3O^{1+}
1-	hydroxide	OH^{1-}
1-	nitrate	NO_3^{1-}
2-	peroxide	O_2^{2-}
3-	phosphate	PO_4^{3-}
2-	sulfate	SO_4^{2-}
2-	sulfite	SO_3^{2-}

Use only for Covalent Compounds

Greek Prefixes

Mono - 1	Hexa – 6
Di – 2	Hepta – 7
Tri – 3	Octa – 8
Tetra – 4	Nona – 9
Penta – 5	Deca – 10

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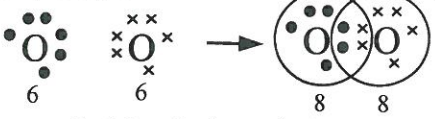
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Covalent Bonding

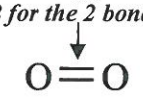
You must fulfill two criteria when making covalent bonds:
1) the individual atoms must have the proper number of valence electrons;
2) when bonded each atom must have 8 electrons through sharing.

Put the number you *need* in the middle to *share*.



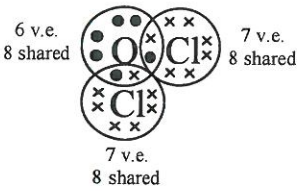
Each has 6 valence electrons by itself and 8 by sharing.

Read each oxygen as 6 v.e. plus 2 for the 2 bonds = 8!

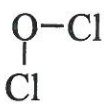


A *double* covalent bond.

Oxygen dichloride: OCl_2



Short hand



Make Cl_2.	Make O_2.	Make P_2.
Make Sulfur difluoride: SF_2	Make sulfur dioxide: SO_2	Make water.

Reviewing

Start getting ready for the test.
Know: valence electrons; oxidation numbers; metals vs. non-metals; dot diagrams; ion notation; cation versus anion; differences between ionic, covalent and polyatomic compounds; how to name compounds; how to make ionic compounds; how to make covalent compounds.

How many protons and electrons does O^{2-} have? Is it a metal or non-metal?	How many electrons does K^{1+} have? Is it a cation or an anion?	Give the ion notation for Calcium that lost 2 electrons. Cation or anion?
Make the ionic compound of Lithium oxide.	Make Iron (III) chloride.	Combine Sodium and phosphate $(PO_4)^{3-}$

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